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To

**Prof. Partha Pratim Roy**

**Pattern Recognition & Visual Computing Lab**

Department of Computer Science & Engineering

Indian Institute of Technology Roorkee

Roorkee, Uttarakhand, India

**Subject: Expression of Interest for Research Internship / Research Trainee Opportunity**

Dear Prof. Roy,

I hope you are doing well.

I am writing to express my sincere interest in the research activities of the Pattern Recognition & Visual Computing Lab, IIT Roorkee and to seek an opportunity to contribute as a Research Intern / Research Trainee under your guidance.

I have recently completed my **Master of Science in Artificial Intelligence and Data Science** from the **Central University of Andhra Pradesh**. My academic background and research interests are **centered on machine learning, deep learning, computer vision, generative AI, and intelligent real-time systems**. During my master's dissertation, I developed a **Distance-Aware Real-Time Assistive Navigation System using YOLOv8 and Monocular Depth Estimation**, where I worked on real-time object detection, spatial reasoning, FastAPI-based deployment, and multilingual feedback mechanisms.

I have also gained practical experience through internships in AI and data science, involving the development of machine learning models, computer vision pipelines, and AI-enabled applications using **Python, Scikit-learn, OpenCV, YOLOv8, FastAPI, and Generative AI tools**. In addition, I have contributed to a peer-reviewed publication related to computer vision and intelligent navigation systems, which strengthened my interest in research methodology, experimentation, and scientific problem solving.

I am particularly motivated by the research environment at IIT Roorkee and would be honored to contribute to ongoing projects involving Computer Vision, Pattern Recognition, Document Analysis, Assistive Visual AI. My background in machine learning models, computer vision pipelines, and system integration aligns closely with your laboratory's focus, and I am eager to apply my technical skills to support your ongoing research initiatives.

I have attached my resume for your kind consideration. I would be grateful if my profile could be considered for any current or upcoming research opportunities in your laboratory.

Thank you very much for your time and consideration. I look forward to the possibility of working with your research group.

Sincerely,

**Adepu Nikhil**